

Physics Beyond the Stars: TALA Hybrid Rocket Launch Set in Capas

CAPAS, TARLAC — The Philippine Space Agency (PhilSA), alongside aerospace engineers and homegrown student researchers, will launch the TALA Hybrid Rocket at the Crow Valley Gunnery Range in Capas, Tarlac. Beyond marking a major milestone in Philippine space technology, the rocket launch provides a clear, real-world demonstration of fundamental physics—specifically Newton’s Laws of Motion, thermodynamics, and aerodynamics.

The fundamental force behind the rocket launch relies directly on Newton’s Third Law of Motion: for every action, there is an equal and opposite reaction. Inside the hybrid rocket engine, solid fuel burns using a liquid oxidizer, generating high-pressure gas. As the burning gas expands and shoots rapidly downward out of the engine nozzle (the action force), it creates a reactive upward push called thrust (the reaction force), accelerating the rocket into the sky.

As the rocket ascends toward its target altitude of 5 kilometers, Newton’s Second Law of Motion dictates its trajectory. The engine’s total force must overcome both gravity pulling the rocket downward and the rocket’s overall mass. Because the rocket burns up propellant as it flies, its mass continuously decreases—allowing its acceleration to increase significantly as it shoots upward through the atmosphere.

Aerodynamics also plays an essential role in keeping the flight stable. The rocket’s sleek nose cone reduces air resistance (drag), while its stabilizing fins distribute aerodynamic forces to keep the rocket traveling on a straight, predictable path.

Once the rocket reaches its highest point, physics continues to govern the recovery phase. The rocket deploys dual parachutes to carry its payload safely back to earth. The parachutes open to drastically increase surface area, creating high air resistance (drag force) that counters gravity and slows the descent to a safe terminal velocity.

The Capas rocket launch is far more than an aerospace milestone. From the high-thrust ignition on the ground to the parachute recovery in the air, every stage of the TALA hybrid rocket flight showcases physics operating in real-world technology—proving that physical science is key to reaching new heights.